TRITON ROBOTICS

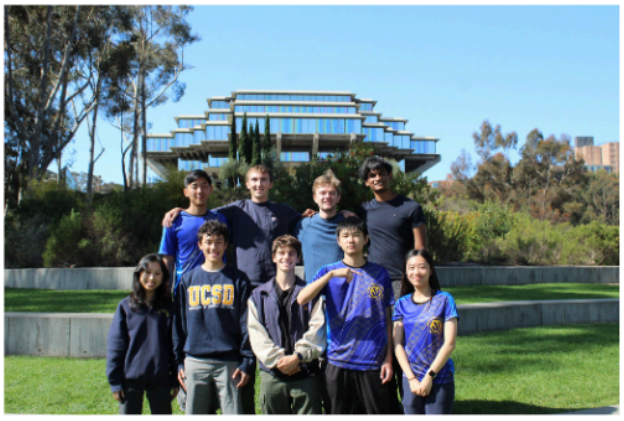
5 Subteams - 3 Robots

Mechanical

Responsible for designing, building, and refining the physical systems of our robots. We use Onshape to prototype mechanisms, collaborate closely with the software and electrical teams to integrate components such as cameras, LiDAR, IMUs, and sliprings, and then bring designs to life through manufacturing techniques like 3D printing, carbon fiber layups, laser cutting, water jetting, and sheet metal fabrication. We continually diagnose issues, machine replacement parts, and implement design improvements to keep our robots reliable and competition-ready.

Electronics and Embedded

Bridges hardware and software to ensure every robot operates safely, reliably, and within competition rules. We integrate sensors and electronics such as IMUs, current sensors, motor controllers, and design PCBs. Our members support autonomy with seamless integration of other peripherals. We also perform iterative testing, debugging, and occasional emergency repairs, making sure every electronic system is competition-ready and tightly synchronized with mechanical and software teams.



Autonomy

Focuses on giving our robots the ability to perceive, localize, and move intelligently in competition. We build computer vision pipelines to detect and track armor panels using classical CV techniques, light-bar filtering, and SolvePNP, enabling accurate autonomous aiming. Our team also develops localization and navigation systems using LiDAR, SLAM, trajectory generation, and path-following algorithms. We maintain a full simulation environment with physics, camera, and LiDAR emulation, and we handle essential dev-ops and system-level management for Jetson hardware. We enable fully autonomous shooting, movement, and decision-making on the field.

Non-Technical/Business

The Non-Technical subteams support Triton Robotics by strengthening our brand, partnerships, and online presence. Marketing & Design creates graphics for social media, merchandise, and outreach materials that visually communicate our work. Business builds relationships with sponsors and industry partners, helping secure funding and opportunities for our members. The Webmaster team maintains and updates our website to showcase projects, events, and achievements. Together, these groups ensure our organization is well-represented, well-supported, and connected to the broader community.